

NE 470/570 Monte Carlo Methods for Radiation Transport

Topic 4: Monte Carlo Integrals and Variance Reduction Techniques

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Outline

1. Monte Carlo Evaluation of Integrals
2. Importance Sampling
3. Control Variates
4. Stratified Sampling
5. Combined Sampling

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1. Monte Carlo Evaluation of Integrals

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- One of the most important applications of the Monte Carlo method is **estimation of integrals of functions or physical quantities**.
- The method is highly flexible, and generally used for solving complex high-dimensional integrals or determining the outcomes of complex physical processes.
- A major issue is the need for **significant computation time** for achieving an acceptable precision or uncertainty.
- Therefore, over the past several decades, significant efforts have been allocated to the development of **variance reduction approaches** that result in **smaller variance in a shorter computation time**.
- In this topic, we will introduce the fundamental concept of Monte Carlo integration, and several representative variance reduction methodologies often applied to particle transport problems.
- The performance of these methodologies is examined based on comparing their variance to that of the standard methodology.

- To see how the Monte Carlo method can be used to estimate a finite integral, we first revisit the definition of expected value of a continuous RV X with PDF $f(x)$

$$I = \mathbb{E}[X] = \int_a^b x f(x) dx, \quad a \leq x \leq b \quad (1)$$

- The expected value of an arbitrary function of X , $g(X)$, is

$$I = \mathbb{E}[g(X)] = \int_a^b g(x) f(x) dx, \quad a \leq x \leq b \quad (2)$$

- Hence, a finite integral is equivalent to the expected value of its integrand (or part of its integrand).
- For example, to evaluate the integrals above, we may sample X based on $f(x)$ and evaluate the mean value as:

$$I_N = \bar{x} = \frac{1}{N} \sum_{i=1}^N x_i \quad (3)$$

and the average $g(X)$ as

$$I_N = \overline{g(x)} = \frac{1}{N} \sum_{i=1}^N g(x_i) \quad (4)$$

- In general, any integral can be evaluated via a Monte Carlo simulation after a PDF is identified for sampling the integration variable.
- For example, consider the following integral

$$I = \int_a^b h(x)dx, \quad a \leq x \leq b \quad (5)$$

Here, it is necessary to identify $g(x)$ (the function being averaged), and $f(x)$ (the PDF) such that $h(x) = g(x)f(x)$.

- The simplest thing is to sample from a uniform distribution given by:

$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b \quad \rightarrow \quad g(x) = \frac{h(x)}{f(x)} = (b-a) \cdot h(x) \quad (6)$$

- To sample from $f(x)$, we form the corresponding FFMC given by:

$$\int_a^x \frac{1}{b-a} dt = \eta \quad \rightarrow \quad x_i = a + \eta_i(b-a) \quad (7)$$

- Then, the integral is calculated by

$$I_N = \overline{g(x)} = \frac{1}{N} \sum_{i=1}^N (b-a)h(x_i) \quad (8)$$

- There are several techniques for variance reduction when determining integrals using Monte Carlo. These techniques are based on modification of different elements of an integral including the **PDF**, the **integrand**, the **domain of integration**, and a **combination of these elements**.
- We will introduce **importance sampling**, **control variates**, **stratified sampling**, and **combined sampling** techniques.
- We need a metric for comparing the performance of different variance reduction techniques. A commonly used metric is the **figure of merit (FOM)** given by:

$$\text{FOM} = \frac{1}{R_x^2 \cdot T} \quad (9)$$

where R_x^2 is the estimated relative uncertainty, and T is the computation time.

- A more effective variance reduction technique should have **higher FOM**, i.e., the technique can achieve a smaller variance in a shorter time.
- It should be noted that the FOM for the same problem and same method will be different on different computers, i.e., it can only be used for **comparison of different methods for the same problem on the same computer**. In the next, we will only examine the effectiveness of each technique **based on the reduction of R_x^2** .

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- **Importance sampling** is based on **modifying the PDF**, in order to achieve a lower variance at a low cost.
- To identify the most effective PDF, we will introduce a **new PDF** $f^*(x)$

$$I = \int_a^b f^*(x) \left[\frac{g(x)f(x)}{f^*(x)} \right] dx, \quad a \leq x \leq b \quad (10)$$

where $f^*(x) \geq 0$, $\int_a^b f^*(x)dx = 1$, and $\frac{g(x)f(x)}{f^*(x)} < \infty$ for $x \in [a, b]$ except at a discrete set of points.

- In order to obtain a formulation for $f^*(x)$, we form the variance of the integrand and try to minimize the variance given by

$$\text{Var}[I] = \int_a^b f^*(x) \left[\frac{g(x)f(x)}{f^*(x)} \right]^2 dx - I^2 \quad (11)$$

- Because the value of I is fixed irrespective of $f^*(x)$, we have to minimize the first term in the above equation, while **maintaining the constraint of $\int_a^b f^*(x)dx = 1$.**

- To do so, we form the corresponding **Lagrange multiplier** expression given by

$$L[f^*(x)] = \int_a^b \frac{g^2(x)f^2(x)}{f^*(x)} dx + \lambda \int_a^b f^*(x) dx \quad (12)$$

and solve for $f^*(x)$ by minimizing the above equation with respect to $f^*(x)$:

$$\frac{\partial L[f^*(x)]}{\partial f^*(x)} = 0 \quad (13)$$

- We need to use the Leibnitz rule:

$$\frac{d}{d\alpha} \left[\int_{h(\alpha)}^{k(\alpha)} f(x, \alpha) dx \right] = \int_{h(\alpha)}^{k(\alpha)} \frac{\partial f(x, \alpha)}{\partial \alpha} d\alpha + f(k(\alpha), \alpha) \frac{dk(\alpha)}{d\alpha} - f(h(\alpha), \alpha) \frac{dh(\alpha)}{d\alpha} \quad (14)$$

- We have:

$$\frac{\partial L[f^*(x)]}{\partial f^*(x)} = - \int_a^b \frac{g^2(x)f^2(x)}{(f^*(x))^2} dx + \lambda \int_a^b dx = 0 \quad (15)$$

$$\int_a^b \left[\frac{g^2(x)f^2(x)}{(f^*(x))^2} - \lambda \right] dx = 0 \quad (16)$$

- To satisfy this equality for any x , it is necessary that the integrand to be equal to zero, i.e.,

$$\frac{g^2(x)f^2(x)}{(f^*(x))^2} - \lambda = 0 \quad (17)$$

- Hence, $f^*(x)$ formulation is given by

$$f^*(x) = \frac{|g(x)f(x)|}{\sqrt{\lambda}} \quad (18)$$

- The value of λ can be determined by requiring that $f^*(x)$ is normalized

$$\int_a^b \frac{g(x)f(x)}{\sqrt{\lambda}} dx = 1 \quad \rightarrow \quad \sqrt{\lambda} = I \quad (19)$$

- Therefore, $f^*(x)$ is given by

$$f^*(x) = \frac{|g(x)f(x)|}{I} \quad (20)$$

- However, the above equation cannot be used in practice because I is unknown. But it indicates that the “best” PDF $f^*(x)$ should be related to the integrand $[g(x)f(x)]$.

Example

- Consider the following integral

$$I = \int_1^2 \ln x dx \quad (21)$$

with an **exact solution** as

$$I = [x \ln x - x] \Big|_1^2 = 0.386294 \quad (22)$$

- A straightforward Monte Carlo simulation will sample x from a uniform distribution in the range of $[1, 2]$. This means that the corresponding FPMC is expressed by

$$x = \eta + 1 \quad (23)$$

- Therefore, the value of the integral after N samples using above equation is given by

$$I_N = \frac{1}{N} \sum_{i=1}^N \ln x_i \quad (24)$$

- The **analytical variance of the integral** (I) is expressed by

$$\text{Var}[I] = \int_1^2 \ln^2 x dx - I^2 = \left[(x \ln^2 x - 2x \ln x + 2x) - (x \ln x - x)^2 \right] \Big|_1^2 = 0.039094 \quad (25)$$

- To obtain a more effective PDF $f^*(x)$, we expand $\ln x$ with the Taylor series expansion as

$$\ln x = \sum_{n=1}^{+\infty} \frac{(-1)^{n+1}}{n} (x-1)^n \quad (26)$$

- For $n=1$, $\ln x \approx x-1$, one may consider a more effective PDF as

$$f^*(x) = \alpha x \quad (27)$$

- Considering that $f^*(x)$ has to be normalized, α is determined by

$$\int_1^2 \alpha x dx = 1 \quad \rightarrow \quad \alpha = \frac{2}{3} \quad (28)$$

- Therefore

$$f^*(x) = \frac{2}{3}x, \quad x \in [1, 2] \quad (29)$$

- The new FPMC and its analytical inversion is:

$$\eta = \int_1^x f^*(t) dt \quad \rightarrow \quad \eta = \frac{x^2 - 1}{3} \quad \rightarrow \quad x = \sqrt{3\eta + 1} \quad (30)$$

- The new integrand $\left[\frac{g(x)f(x)}{f^*(x)} \right]$ is expressed by

$$g^*(x) = \frac{\ln x}{\frac{2}{3}x} \quad (31)$$

- Then, the integral is determined by

$$I_N^* = \frac{1}{N} \sum_{i=1}^N \frac{\ln x_i}{\frac{2}{3}x_i} \quad (32)$$

where x_i is sampled from $f^*(x)$.

- The **theoretical variance for the revised integrand** is given by

$$\text{Var}[I] = \int_1^2 \left[\frac{\ln x}{\frac{2}{3}x} \right]^2 \left(\frac{2}{3}x \right) dx - I^2 = \left[0.5 \ln^3 x \right]_1^2 - I^2 = 0.017289 \quad (33)$$

- Hence, by considering only the first term of the $\ln x$ expansion, may achieve faster convergence by a factor ≈ 2 .

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- Instead of modifying the PDF $f(x)$, we can **change the function $g(x)$** such that $\text{Var}[I]$ is reduced. We can subtract and add a function $h(x)$ as follows

$$I = \int_a^b f(x)g(x)dx = \int_a^b f(x) [g(x) - h(x)] dx + \int_a^b f(x)h(x)dx \quad (34)$$

- To achieve a lower variance, $h(x)$ has to have the following properties:

- 1 **it should be similar to $g(x)$;**
- 2 **the second integral has an analytical solution, I_h .**

- This means that to evaluate I , we only need to sample the first term, i.e.,

$$I_N = \frac{1}{N} \sum_{i=1}^N [g(x_i) - h(x_i)] + I_h \quad (35)$$

- If the difference $(g(x) - h(x))$ is almost a constant for all x_i 's, then it is expected that $\text{Var}[g(x) - h(x)] \leq \text{Var}[g(x)]$.

Example

- Let's still use the example $I = \int_1^2 \ln x dx$, where $g(x) = \ln x$, and $f(x) = 1$.
- Consider $h(x) = x$, $\text{Var}[g(x) - h(x)]$ is given by

$$\text{Var}[g(x) - h(x)] = \int_1^2 (\ln x - x)^2 dx - \left[\int_1^2 (\ln x - x) dx \right]^2 \quad (36)$$

- The first integral in the above equation is given by

$$\int_1^2 (\ln x - x)^2 dx = \left[x \ln^2 x - (2x + x^2) \ln x + 2x + \frac{x^2}{2} + \frac{x^3}{3} \right] \Big|_1^2 = 1.24906 \quad (37)$$

Example

- And the the second integral is given by

$$\left[\int_1^2 (\ln x - x) dx \right]^2 = \left[\left(x \ln x - x - \frac{x^2}{2} \right) \Big|_1^2 \right]^2 = 1.24034 \quad (38)$$

- Therefore, $\text{Var}[g(x) - h(x)] = 0.008722$.
- For this example, the control variates technique may result in a speedup of ≈ 4.5 , which is twice as fast as importance sampling.
- In short, both the important sampling and control variates techniques have resulted in speedup with relatively minimal changes to PDF and/or integrand.

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- The **stratified sampling** technique examines and partitions the domain of integration, and sets the amount of sampling for each subdomain according to its importance or need for achieving acceptable precision. It has been used in different applications.
- One of the common applications of stratification is **population survey**. Rather than doing a simple random sampling, one identifies “important” variables and establishes “homogeneous” subgroups (referred to as **strata**) based on these variables, then each subgroup is sampled according to its proportion in the population.
- If appropriate variables are selected, this method will have a higher precision over simple random sampling because it focuses on important subgroups and avoids “unrepresentative” sampling.

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- If appropriate variables are selected, this method will have a higher precision over simple random sampling because it focuses on important subgroups and avoids “unrepresentative” sampling.
- **Stratified sampling also enables a smaller sample size**. For example, for a population of size N partitioned into M subgroups with N_m population, if we consider a sample size of $n \ll N$, the proportionate sample sizes of different subgroups (strata) can be calculated by

$$n_m = \frac{N_m}{N} \cdot n, \quad m = 1, 2, \dots, M \quad (39)$$

- **Since with stratified sampling an appropriate number of samples is used for each stratum, it is possible to achieve a higher precision with a smaller sample size.**

- Another important application of this sampling approach is the **evaluation of mathematical integrals**. First, we discuss in detail the implementation process of stratified sampling, then we derive the formulation for selecting appropriate sample size that can lead to reduced variance.
- As an example, consider the integral

$$I = \int_a^b g(x)f(x)dx \quad (40)$$

- We partition the integral domain $D = [a, b]$ into M non-overlapping subdomains (strata)

$$\{D_m\}_{m=1}^M \quad (41)$$

such that

$$I = \sum_{m=1}^M \int_{D_m} g(x)f(x)dx \quad (42)$$

- To proceed, define the PDF for sampling x for any subdomain D_m as

$$f_m(x) = \frac{f(x)}{\int_{D_m} f(x)dx} \quad \text{such that} \quad \int_{D_m} f_m(x)dx = \int_{D_m} \frac{f(x)}{\int_{D_m} f(x)dx} dx = 1 \quad (43)$$

- Consider a formulation for the **fractional area of each subdomain** as

$$h_m = \frac{\int_{D_m} f(x) dx}{\int_D f(x) dx} = \int_{D_m} f(x) dx \quad (44)$$

- Then we have:

$$f(x) = h_m \cdot f_m(x), \quad m = 1, 2, \dots, M \quad (45)$$

- The integral I becomes

$$I = \sum_{m=1}^M \int_{D_m} g(x) f(x) dx = \sum_{m=1}^M h_m \int_{D_m} g(x) f_m(x) dx \quad (46)$$

- Now, applying the Monte Carlo method for evaluating the subdomain integrals, the **stratification formulation for integral (I_s)** is given by

$$I_s = \sum_{m=1}^M \frac{h_m}{N_m} \sum_{i=1}^{N_m} g_m(x_i) \quad (47)$$

where N_m is the number of samples obtained for subdomain D_m using the PDF $f_m(x)$.

- Still, the ideal number of samples per subdomain (N_m) is not known; therefore, it is necessary to develop a formulation for (N_m).
- To do so, we need to minimize the variance of I_s , while preserving the total number of samples, i.e., $N = \sum_{m=1}^M N_m$.
- The variance I_s is given by

$$\begin{aligned}\sigma^2(I_s) &= \sum_{m=1}^M \frac{h_m^2}{N_m^2} \sum_{i=1}^{N_m} \sigma^2(g_m(x_i)) \\ &= \sum_{m=1}^M \frac{h_m^2}{N_m^2} N_m \sigma_m^2 = \sum_{m=1}^M \frac{h_m^2}{N_m} \sigma_m^2\end{aligned}\tag{48}$$

where $\sigma_m^2 = \sigma^2(g_m)$ in the m^{th} subdomain (strata).

- To minimize the variance, while maintaining a constraint on the total sample size, we use the Lagrange multiplier formulation expressed by

$$L[N_m] = \sum_{m=1}^M \frac{h_m^2}{N_m} \sigma_m^2 + \lambda \sum_{m=1}^M N_m\tag{49}$$

- Then, we minimize the Lagrange multiplier by finding its derivative with respect to N_m

$$\frac{\partial L[N_m]}{\partial N_m} = \sum_{m=1}^M \left[\frac{h_m^2}{N_m^2} \sigma_m^2 - \lambda \right] = 0 \quad (50)$$

- To satisfy the above equality for any m , we set the term inside the bracket to be zero, and obtain a formulation for λ in terms of N_m

$$\sqrt{\lambda} = \frac{h_m \sigma_m}{N_m} \quad (51)$$

- Then, we use the sample size constraint to obtain another formulation for $\sqrt{\lambda}$

$$N = \sum_{m=1}^M N_m = \frac{1}{\sqrt{\lambda}} \sum_{m=1}^M h_m \sigma_m \rightarrow \sqrt{\lambda} = \frac{\sum_{m=1}^M h_m \sigma_m}{N} \quad (52)$$

- If we equate the two formulations for $\sqrt{\lambda}$, we can derive a formulation for the sample size for each subdomain (D_m) as

$$N_m = \frac{h_m \sigma_m}{\sum_{m=1}^M h_m \sigma_m} \cdot N \quad (53)$$

- The formulation for N_m indicates that the number of samples per subdomain is weighted by the **combined effects of fractional area and variance of each subdomain**.
- In other words, **a subdomain with a larger combined area and variance requires more samples in order to obtain a smaller variance** for a given total sample size N .
- Of course, this formulation is not useful unless one estimates the variances of all the subdomains. This can be achieved by determining variances based on a relatively small number of sampling.
- We replace N_m in the relation of $\sigma^2(I_s)$, then the variance formulation of I_s is expressed by

$$\begin{aligned}\sigma^2(I_s) &= \sum_{m=1}^M \frac{h_m^2}{N_m} \sigma_m^2 = \sum_{m=1}^M \frac{h_m^2}{\frac{h_m \sigma_m}{\sum_{m=1}^M h_m \sigma_m} \cdot N} \sigma_m^2 \\ &= \frac{1}{N} \left[\sum_{m=1}^M h_m \sigma_m \right]^2 = \frac{1}{N} (\overline{\sigma_s})^2\end{aligned}\tag{54}$$

where $\overline{\sigma_s}$ is given by

$$\overline{\sigma_s} = \sum_{m=1}^M h_m \sigma_m\tag{55}$$

- To examine the effectiveness of the stratification approach, we have to compare the above variance to the variance of the **standard Monte Carlo formulation** given by

$$I_N = \int_D f(x)g(x)dx = \bar{g} = \frac{1}{N} \sum_{i=1}^N g(x_i) \quad (56)$$

- The variance of I_N is expressed by

$$\sigma^2(I_N) = \sigma^2 \left[\frac{1}{N} \sum_{i=1}^N g(x_i) \right] = \frac{1}{N^2} \sum_{i=1}^N \sigma^2(g(x_i)) = \frac{1}{N} \sigma_D^2 \quad (57)$$

where

$$\sigma_D^2 = \int_D f(x) (g(x) - \bar{g})^2 dx \quad (58)$$

- However, this formula for $\sigma^2(I_N)$ cannot be directly compared to $\sigma^2(I_s)$.

- To make a more straightforward comparison between $\sigma^2(I_N)$ and $\sigma^2(I_s)$, we use the stratified sampling partition $\{D_m\}_{m=1}^M$ to evaluate $\sigma^2(I_N)$, but without using its optimized sample sizes N_m .

$$\sigma^2(I_N) = \frac{1}{N} \sum_{m=1}^M h_m \int_{D_m} f_m(x) (g(x) - \bar{g})^2 dx \quad (59)$$

- Add and subtract $\overline{g_m} = \int_{D_m} f_m(x)g(x)dx$ in $(g(x) - \bar{g})$

$$\begin{aligned} \sigma^2(I_N) &= \frac{1}{N} \sum_{m=1}^M h_m \int_{D_m} f_m(x) (g(x) - \overline{g_m} + \overline{g_m} - \bar{g})^2 dx \\ &= \frac{1}{N} \sum_{m=1}^M h_m \int_{D_m} f_m(x) \left[(g(x) - \overline{g_m})^2 + (\overline{g_m} - \bar{g})^2 + 2(g(x) - \overline{g_m}) \cdot (\overline{g_m} - \bar{g}) \right] \\ &= \frac{1}{N} \sum_{m=1}^M h_m \sigma_m^2 + \frac{1}{N} \sum_{m=1}^M h_m (\overline{g_m} - \bar{g})^2 \int_{D_m} f_m(x) dx \\ &\quad + \frac{2}{N} \sum_{m=1}^M h_m (\overline{g_m} - \bar{g}) \left[\int_{D_m} f_m(x)g(x)dx - \overline{g_m} \int_{D_m} f_m(x)dx \right] \end{aligned}$$

- Because

$$\int_{D_m} f_m(x) dx = 1 \quad \text{and} \quad \int_{D_m} f_m(x) g(x) dx = \overline{g_m} \quad (60)$$

- The equation for $\sigma^2(I_N)$ reduces to

$$\begin{aligned} \sigma^2(I_N) &= \frac{1}{N} \sum_{m=1}^M h_m \sigma_m^2 + \frac{1}{N} \sum_{m=1}^M h_m (\overline{g_m} - \bar{g})^2 + \frac{2}{N} \sum_{m=1}^M h_m (\overline{g_m} - \bar{g}) [\overline{g_m} - \overline{g_m}] \\ &= \frac{1}{N} \sum_{m=1}^M h_m \sigma_m^2 + \frac{1}{N} \sum_{m=1}^M h_m (\overline{g_m} - \bar{g})^2 \end{aligned} \quad (61)$$

- To evaluate the effectiveness of the stratified sampling technique, the difference Δ between the standard variance $\sigma^2(I_N)$ and the stratified sampling variance $\sigma^2(I_s)$ is

$$\begin{aligned} \Delta &= \sigma^2(I_N) - \sigma^2(I_s) \\ &= \frac{1}{N} \sum_{m=1}^M h_m \sigma_m^2 + \frac{1}{N} \sum_{m=1}^M h_m (\overline{g_m} - \bar{g})^2 - \frac{1}{N} (\overline{\sigma_s})^2 \\ &= \frac{1}{N} \sum_{m=1}^M h_m \sigma_m^2 + \frac{1}{N} \sum_{m=1}^M h_m (\overline{g_m} - \bar{g})^2 - \frac{1}{N} \left(\sum_{m=1}^M h_m \sigma_m \right)^2 \end{aligned} \quad (62)$$

- To simplify the Equation (62), we first prove an inequality:

$$\begin{aligned}
 \frac{1}{N} \sum_{m=1}^M h_m (\sigma_m - \bar{\sigma}_s)^2 &= \frac{1}{N} \sum_{m=1}^M h_m (\sigma_m^2 + (\bar{\sigma}_s)^2 - 2\sigma_m \bar{\sigma}_s) \\
 &= \frac{1}{N} \sum_{m=1}^M h_m \sigma_m^2 + \frac{1}{N} \sum_{m=1}^M h_m (\bar{\sigma}_s)^2 - 2 \frac{1}{N} \sum_{m=1}^M h_m \sigma_m \bar{\sigma}_s \\
 &= \frac{1}{N} \sum_{m=1}^M h_m \sigma_m^2 + \frac{1}{N} (\bar{\sigma}_s)^2 - 2 \frac{1}{N} (\bar{\sigma}_s)^2 \\
 &= \frac{1}{N} \sum_{m=1}^M h_m \sigma_m^2 - \frac{1}{N} (\bar{\sigma}_s)^2
 \end{aligned} \tag{63}$$

- Therefore, we have:

$$\Delta = \frac{1}{N} \sum_{m=1}^M h_m (\sigma_m - \bar{\sigma}_s)^2 + \frac{1}{N} \sum_{m=1}^M h_m (\bar{g}_m - \bar{g})^2 > 0 \tag{64}$$

- The above formulation indicates that the difference Δ is always positive, so $\sigma^2(I_N) > \sigma^2(I_s)$. We have proven that using an optimum set of N_m 's, **it is guaranteed that the integral variance when using stratified sampling is always smaller than that achieved by the standard technique.**

Example

- Consider the following integral:

$$I = \int_{-3}^3 (1 + \tanh x) dx \quad (65)$$

- Given $g(x)f(x) = (1 + \tanh x)$, using a uniform distribution defined on $[-3, 3]$, we have $f(x) = \frac{1}{6}$. The integral becomes

$$I = 6 \cdot \int_{-3}^3 \frac{1 + \tanh x}{6} dx \quad (66)$$

- We derive the formulations for $f_m(x)$ and h_m , with D_m (subdomain interval) equal to 1.
- The PDF for each subdomain $f_m(x)$ and its fractional area h_m are:

$$f_m(x) = \frac{f(x)}{\int_{D_m} f(x) dx} = \frac{\frac{1}{6}}{\int_{D_m} \frac{1}{6} dx} = 1 \quad (67)$$

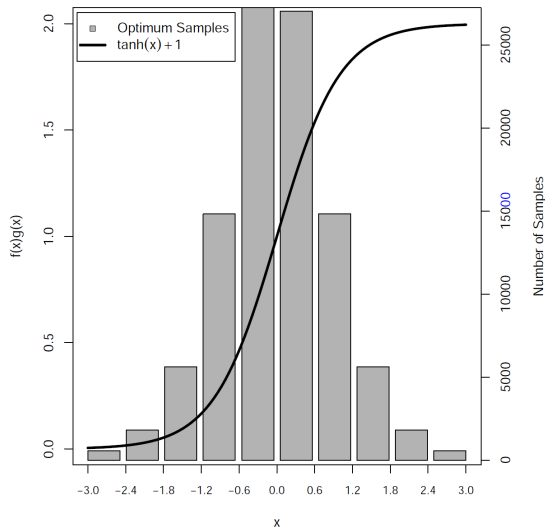
$$h_m = \int_{D_m} f(x) dx = \int_{D_m} \frac{1}{6} dx = \frac{1}{6} \quad (68)$$

- Three distributions of number of samples are considered including “**standard**” (randomly from a uniform distribution), “**uniform**” (with equal number of samples per stratum), and “**optimum**”.
- To generate an estimate of “optimum” number of samples per stratum, we first consider two cases with 100 and 1,000 total number of samples N , divide the range of integral into 10 strata, and estimate the variance per stratum.
- Table 5.1 compares the estimated optimum number of samples per strata to those obtained using the standard approach for the two cases.

Table 5.1 Comparison of number of samples per strata for the “standard” and “optimum” cases for 100 and 1000 sampled

Case	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
100 Samples										
standard	10	10	14	6	11	13	5	12	8	11
optimum	2	2	2	16	25	24	12	7	3	2
1000 samples										
standard	106	103	88	104	98	99	111	89	103	99
optimum	7	18	55	155	271	266	150	53	19	6

- As expected the estimated number of samples (N_m) is highly variable in the 100-sample case as compared to the 1,000-sample case, which is approaching an expected uniform distribution.
- This, however, is not true for the optimum distribution, which has redistributed the number of samples based on the larger or smaller variance per strata.
- The right figure overlays the distribution of the integrand and the optimum distribution of number of samples.
- It indicates that the formulation for the optimum number of samples per strata indeed has assigned significantly larger number of samples for the middle segment of integrand that exhibits large variation versus the two tails which exhibit smooth distributions.

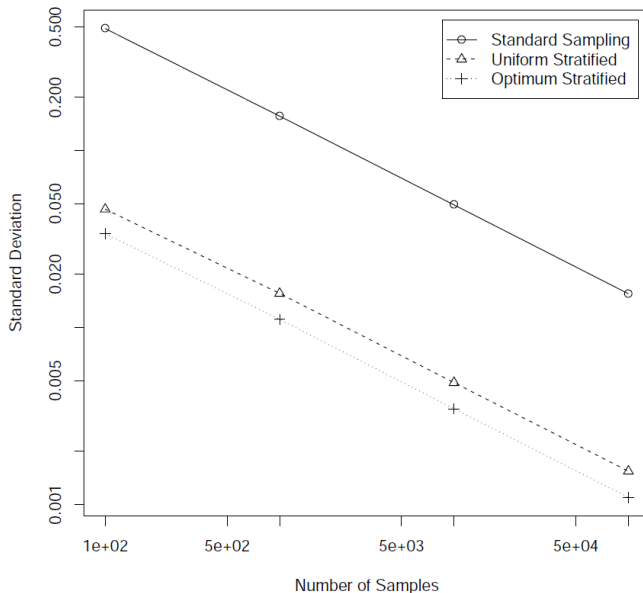


- To examine the effectiveness of the stratified sampling, we compare the analytical value of the integral to those estimated via the standard and the stratified sampling approaches.
- Table 5.2 compares the estimated integrals and associated standard deviations and relative differences for increasing number of samples.
- It indicates that stratified sampling even with equal number of samples per strata yields significantly lower standard deviation and relative uncertainty as compared to the standard approach.
- The optimum stratified sampling performs the best. The estimated integral is in excellent agreement with the analytical value, especially for the optimum sampling approach that agrees within the estimated uncertainty.

Table 5.2 Comparison of the predictions of the standard and stratified sampling techniques to the analytical solutions (Equation 5.69)

Method	Number of samples	Expected value	sample S_x	Rel. Diff. (%)
Standard				
	100	5.97237	0.50884	8.52
	1,000	5.94319	0.15494	2.61
	10,000	5.96752	0.04928	0.83
	100,000	5.99457	0.01555	0.26
Uniform stratified				
	100	5.97234	0.05588	0.94
	1,000	5.97917	0.01480	0.25
	10,000	5.99955	0.00492	0.08
	100,000	6.00088	0.00154	0.03
Optimum stratified				
	100	5.96951	0.03421	0.57
	1,000	6.01209	0.01104	0.18
	10,000	6.00326	0.00343	0.06
	100,000	6.00135	0.00109	0.02

- The figure shows the expected behavior of the standard deviation for different sampling approaches with the number of samples as given in Table 5.2.
- Theoretically, if the solution has converged to where the central limit theorem is valid, then the variance drops by the ratio of square roots of numbers of samples.
- For the current example, this ratio is $\frac{1}{\sqrt{10}}$ between two consecutive cases from 100 to 100,000.



- To make sense of the effect of the uniform stratification, we look at the follow relation based on Equation (64)

$$\Delta = \sigma^2(I_N) - \sigma^2(I_s) = \frac{1}{N} \sum_{m=1}^M h_m (\sigma_m - \bar{\sigma}_s)^2 + \frac{1}{N} \sum_{m=1}^M h_m (\bar{g}_m - \bar{g})^2 \quad (69)$$

so we have

$$\sigma^2(I_N) = \frac{1}{N} \sum_{m=1}^M h_m (\sigma_m - \bar{\sigma}_s)^2 + \frac{1}{N} \sum_{m=1}^M h_m (\bar{g}_m - \bar{g})^2 + \sigma^2(I_s) \quad (70)$$

- With the **uniform stratified sampling**, the second term is eliminated. With the **optimum stratified sampling**, the first term is also eliminated, leaving only the third term.
- In this example, the uniform stratified sampling performed almost as well as the optimum case, indicating that the second term was higher than the first term.
- The relative importance of these terms can vary significantly for different applications, and therefore impacting the effectiveness of the stratified sampling techniques.

Outline

1. Monte Carlo Evaluation of Integrals
2. Importance Sampling
3. Control Variates
4. Stratified Sampling
5. Combined Sampling

- In principle, depending on the application, one can develop new techniques by combining the aforementioned variance reduction techniques.
- For example, one can combine importance sampling and stratified sampling.
- This means that after the $f^*(x)$ is identified, a new integrand $g^\dagger(x)$ is formed, then the behavior of the integrand is considered for partitioning the domain into M subdomains.
- The integral equation for importance sampling

$$I = \int_a^b f^*(x) \left[\frac{g(x)f(x)}{f^*(x)} \right] dx, \quad a \leq x \leq b \quad (71)$$

can be rewritten as

$$I = \sum_{m=1}^M h_m \int_{D_m} g^\dagger(x) f_m^*(x) dx \quad (72)$$

where

$$g^\dagger(x) = \frac{g(x)f(x)}{f^*(x)} \quad \text{and} \quad f_m^*(x) = \frac{f^*(x)}{h_m} \quad (73)$$

- Then, for each subdomain, $f_m^*(x)$ is sampled to obtain the average value of integrand per subdomain as

$$\overline{g_m^\dagger} = \frac{1}{N_m} \sum_{i=1}^{N_m} g_m^\dagger(x_i) \quad (74)$$

- Hence, the integral is given by

$$I = \sum_{m=1}^M h_m \overline{g_m^\dagger} \quad (75)$$

- This method should yield a more accurate solution given that an appropriate importance function and sample size per subdomain are selected. Note that sample sizes have to be obtained based on the the following relation

$$N_m = \frac{h_m \sigma_m}{\sum_{m=1}^M h_m \sigma_m} \cdot N \quad (76)$$

- In addition to minimization of variance, it is important to make sure the computation time is also reduced, i.e., ultimately, the goal is to achieve high FOM.